

KUNDAN SRINIVAS SAKKURU

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MS Computer Science — University of South Florida, May 2026

SUMMARY

Full-stack software engineer specializing in production AI systems, cross-platform mobile, and scalable web applications. Hands-on experience building and shipping generative AI pipelines, agentic systems, and real-time data applications used by real users in production. Two-time hackathon award winner. Published researcher with 8 papers in ML, computer vision, and mobile systems. Proven ability to take full end-to-end ownership — from system design and development to deployment and iteration.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript (ES6), Dart, C, SQL, Bash

Web & Mobile: React, Django, HTML, CSS, FastAPI, WebSocket, REST APIs, Flutter, Android SDK (Java), iOS, React Native, Swift

Data & Cloud: Firebase Firestore, Firebase Auth, PostgreSQL, MySQL, MongoDB, SQLite, Android Room, Google OAuth, Python Cloud Functions, GCP, AWS, CI/CD, GitHub Actions

AI & ML: PyTorch, TensorFlow, Keras, Scikit-learn, Generative AI, LangChain, Agentic AI, RAG, Vector Search, ChromaDB, sentence-transformers, LLM Evaluation, Prompt Engineering, Few-shot Learning, Zero-shot Learning, Knowledge Distillation, MediaPipe, OpenCV, Computer Vision, NLTK, Pandas, NumPy, Google Gemini, ElevenLabs, Speech Synthesis

Tools & Practices: Git, GitHub, Postman, Android Studio, Jupyter, VS Code, Agile/Scrum, SDLC, Unit Testing, Code Review, Cross-functional Collaboration

EXPERIENCE

Research Software Engineer

Aug 2025 – May 2026

SHIELD Lab, University of South Florida | Tampa, FL

- Sole engineer behind **CogniX** — a **production-deployed cross-platform Flutter app** (Android and iOS) actively used across live neuroscience studies, administering reaction time, working memory, and sustained attention assessments with **sub-millisecond Dart isolate event logging**; replaced legacy manual processes and became the lab's primary real-user data collection system.
- Designed a **scalable, zero-collision Firestore schema** (participantId → sessionId → trialIndex) supporting high-concurrency multi-device sessions with offline-first persistence and automatic sync on reconnect — achieving **zero data loss across 50+ real production sessions** in low-connectivity field environments.
- Eliminated 100% of manual data entry** by shipping a serverless Python Cloud Function that triggers on session completion and auto-delivers analysis-ready CSV (pandas) and formatted PDF (ReportLab) reports to researchers — turning a multi-hour manual task into an instant automated pipeline.
- Operated in tight **Agile sprints** with neuroscientists; owned requirements gathering, development, testing, and deployment end-to-end — iterating rapidly based on feedback from live participant sessions.

Artificial Intelligence Intern

Jan 2023 – Jun 2023

YoungMinds | Remote

- Built and shipped **production-ready end-to-end ML pipelines** for NLP text classification and predictive analytics — owning the full lifecycle from data ingestion, feature engineering, and model training through evaluation and inference deployment within Agile sprint cycles.
- Drove model quality through rigorous evaluation on precision, recall, F1, and ROC-AUC; applied targeted hyperparameter tuning to reduce overfitting and delivered **data-backed performance reports** to stakeholders every sprint.

PROJECTS

CogniX — Production Neurocognitive Assessment Platform

Flutter | Dart Isolates | Firebase Firestore | Serverless | Python Cloud Functions | pandas | ReportLab | Android | iOS

- Built and **deployed a production-grade cross-platform Flutter app** (Android and iOS) with Dart isolate-based precision timing that captures user response latency at the exact hardware interaction moment — not the UI repaint cycle — ensuring scientifically valid millisecond-accurate data across three neurocognitive test types for real study participants.
- Engineered a **fully automated serverless data pipeline**: concurrent Firestore session writes with offline-first persistence for field reliability, and a Python Cloud Function delivering analysis-ready CSV and PDF reports on session completion — eliminating all manual data handling and giving researchers instant access to structured, publication-quality data.

BayShield — Agentic AI Disaster Response System | HackUSF 2026 — 3rd Place

Python | LangChain | Agentic AI | RAG | Vector Search | ChromaDB | sentence-transformers | FastAPI | WebSocket | React

- Designed and deployed a **production agentic AI system in 24 hours** featuring four autonomous specialized agents running in parallel: real-time NOAA weather data ingestion, shelter-capacity optimization against population density maps, cross-zone resource allocation, and synthesized incident command reports — all operating on live Hurricane Helene disaster data.
- Engineered a **RAG pipeline with vector search** grounding every agent decision in verified historical precedent: FEMA post-incident reports embedded into ChromaDB via sentence-transformers, with semantic similarity retrieval eliminating hallucination risk on high-stakes emergency response calls. Delivered a live React dashboard over a **scalable FastAPI WebSocket backend** with 5-second real-time refresh.

Finderly — Full-Stack Intelligent Matching Web App

React | TypeScript | Firebase Auth | Google OAuth | Firestore | REST APIs | Real-time

- Designed and shipped a **full-stack real-time web application** from scratch featuring a custom TypeScript token-overlap scoring engine that automatically matches and ranks found-item reports against lost-item descriptions using Firestore composite queries across category, location, and date — replacing manual search with intelligent automated matching.
- Delivered **real-time peer-to-peer messaging** using Firestore onSnapshot listeners with per-conversation security rules enforcing strict two-party access, plus multi-provider Firebase Auth (Google OAuth and email/password) — built entirely from first principles with no third-party SDKs for maximum performance and control.

SignBridge — Real-Time Generative AI Sign Language Platform | HackaBull 2025 — Standout Entry, 2nd Place

MediaPipe | Google Gemini | NLTK | Django | ElevenLabs | Speech Synthesis | Computer Vision | Python

- Architected and shipped a **production-ready two-way real-time communication system** in 24 hours targeting 70M+ sign language users worldwide: a Speech-to-Sign pipeline uses NLTK-based NLP grammar restructuring to convert English to grammatically correct ASL; a Sign-to-Speech pipeline applies **MediaPipe computer vision** landmark tracking and **Google Gemini generative AI** to interpret gestures and synthesize natural speech via **ElevenLabs TTS API**.
- Engineered **sub-second end-to-end latency** across both pipelines by solving the hard structural incompatibility between ASL and English grammar through real-time NLP reordering and motion-aware spatial gesture recognition — enabling fully autonomous two-way conversation with zero human interpreter dependency.

Evaluating Distilled Models in In-Context Learning

Python | Generative AI | LLM Evaluation | Prompt Engineering | Hugging Face | NLP Benchmarks

- Conducted **rigorous evaluation of knowledge-distilled generative AI models** under few-shot and zero-shot in-context learning across standard NLP benchmarks — systematically varying prompt structure, shot count, and formatting to quantify how distillation degrades or preserves contextual reasoning in smaller models.
- Produced **actionable model selection guidance** by analyzing trade-offs between model size, inference latency, and task accuracy — directly applicable to production teams choosing between full-scale and distilled LLMs for resource-constrained or latency-sensitive deployment environments.

Skin Cancer Prediction using CNN | Published — Scopus

Python | PyTorch | TensorFlow | Keras | Computer Vision | Deep Learning | NumPy | Pandas | ResNet50

- Designed and trained a **production-quality deep learning CNN** for 7-class skin lesion classification on the HAM10000 clinical dataset — five convolutional blocks with batch normalization and dropout, with performance benchmarked against ResNet50 transfer learning and results **published in a Scopus-indexed journal**.
- Solved a critical **67% class imbalance** through inverse-frequency class weighting and targeted ImageDataGenerator augmentation on underrepresented classes, achieving an **18% improvement in minority-class recall** over the unweighted baseline — demonstrating real-world clinical utility beyond raw accuracy metrics.

EDUCATION

MS, Computer Science

Aug 2024 – May 2026

University of South Florida | Tampa, FL

Coursework: Operating Systems, Algorithms, Machine Learning, Mobile Systems, Software Engineering

BE, Computer Science and Engineering

Nov 2020 – May 2024

R.M.D Engineering College, Anna University | India

CGPA: 9.03/10 | Gold Medal, Best Outgoing Student | Coursework: Compilers, OS, Data Structures, Computer Networks

ACHIEVEMENTS

- **HackaBull 2025, Standout Entry, 2nd Place** — SignBridge: production generative AI sign language platform for 70M+ users, built and demoed in 24 hours.
- **HackUSF 2026, 3rd Place** — BayShield: production agentic AI disaster response system running on real Hurricane Helene data, built in 24 hours.
- **Gold Medal, Best Outgoing Student** — Highest academic performance across entire undergraduate cohort; CGPA 9.03/10.
- **Smart India Hackathon 2022, National Finalist** — Selected among top teams nationally in Ministry of Education innovation challenge.
- **Cognizant PRODIGI Hackathon, Finalist | TNSI Program Phase II, Selected | Centre of Excellence in Cloud Computing, Selected**